Feiyang **Chen**

AI Application Engineer | Agents / RAG / Backend Development

Suzhou, Jiangsu, China

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Summary

AI application engineer with a Master of Computer Science from the University of Sydney and approximately three years of Linux kernel and systems development experience. Independently developed and continue to maintain a news proofreading agent spanning fact extraction/retrieval, structured-output validation, model deployment, and MCP integration; contributed to a news content platform. Experienced in debugging, automated testing, and software delivery, with open-source work in LangGraph, RAG, and offline evaluation.

Skills

Languages	Python, C
AI Applications	LangChain/LangGraph, RAG, MCP
Backend	FastAPI, PostgreSQL, Redis
Engineering	Linux Developing, Cross-architecture Porting, Docker

Projects

News Proofreading and Content Management

Suzhou, Jiangsu, China

Core Developer (Project Collaboration; Development and Maintenance) *Apr. 2026 – Present*

- **AI Proofreading:** Independently owned requirements, architecture, delivery, and ongoing maintenance of a FastAPI/LangChain service for language and factual-consistency checks on news articles.
- Built a pipeline for news ingestion, entity and appointment extraction, and PostgreSQL fact retrieval, with alias matching and evidence ordered by recency; instructed the agent to ground dynamic factual corrections in retrieved evidence.
- Combined Pydantic schemas with source-span and replacement validation; fed errors back to the model for retries and bounded model/tool calls to handle invalid or unlocatable suggestions.
- Built an editor SDK for accepting/rejecting suggestions and applying edits while preserving Markdown/HTML structure; exposed proofreading to WorkBuddy through FastMCP, integrating JWT authentication, rate limits, quota consumption, and refunds for pre-execution failures.
- Deployed Qwen locally and built Taskiq/Redis ingestion, deduplication, caching, and Langfuse tracing; added pytest/Testcontainers and Vitest tests for retrieval, output validation, deduplication, and edit application.
- **Business System:** Owned administration frontend/backend development and portal integration in a team, covering multi-tenant content management, legacy data migration, third-party synchronization, and publishing schedules.
- **Personal Open-source Validation:** Built an A-share analysis agent with LangGraph routing/query rewriting and BM25/vector retrieval fused by RRF; evaluated faithfulness and tool use with DeepEval and documented architecture choices and lessons learned.

Systems Software and Open-source Contributions

Suzhou, Jiangsu, China

Kernel and Systems Development (Including Open-source Maintenance) *2020 – 2025*

- Added LoongArch64/Linux support to Valgrind, contributing libVEX integer, floating-point, and vector instruction translation and runtime support; reached initial usability in one month, passed hundreds of regression tests, and added platform/nightly test configurations.
- Contributed to Linux system call and network driver porting; added LoongArch32/64 support to nolibc, statx compatibility, and platform selftests.
- Added LoongArch and QEMU platform support to RT-Thread and supported LS2K0300 integration; ported Newlib and added LoongArch support to Kcov.

Work Experience

Loongson Technology Corporation Limited

Suzhou, Jiangsu, China

Kernel Engineer *Dec. 2020 – Aug. 2023*

- Submitted upstream MIPS/LoongArch Linux patches, covering root-cause analysis, fixes, builds, and regression testing.
- Led LoongArch enablement for klibc, Valgrind, and other low-level software; resolved build, instruction, runtime, and performance issues.
- Improved GitHub build, test, and release workflows, increasing delivery reliability and team efficiency.

Jiangsu Lemote Technology Co., Ltd.

Suzhou, Jiangsu, China

Kernel Engineer *Jun. 2020 – Dec. 2020*

- Developed and debugged Linux kernel code through issue reproduction, log analysis, patching, and validation.
- Ported OpenStack Nova and DevStack to Fedora MIPS, resolving component compatibility and environment issues; brought up the core service workflow within one month.

Education

The University of Sydney

Sydney, NSW, Australia

Master of Computer Science *Feb. 2024 - May 2026*

- **ISSRE 2026 accepted paper (second author):** "Dr. DD: 1-Minimal Isolation of Failure Causes via Deferred Restarts."
- **Implementation and Experiments:** Built a Python delta debugging framework; evaluated reduction algorithms on binary inputs triggering Binutils/Valgrind bugs, comparing reduction quality, oracle calls, and execution time.

Suzhou University of Technology (formerly Changshu Institute of Technology)

Suzhou, Jiangsu, China

Bachelor of Computer Science and Technology *Sep. 2016 - Jun. 2020*

- **ICAIS 2020 paper (first author):** "Formal Verification of the Correctness of Operating System on Assembly Layer in Isabelle/HOL."
- **Research:** Modelled MINIX 3's i386 context-saving assembly in Isabelle/HOL and verified register/memory states and instruction execution against expected results.